

INVESTHUB

DBMS PROJECT

**GIVEN BY
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GROUP MEMBERS

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INTRODUCTION

We are developing a database project on the **Stock Market Management System**, named **InvestHUB**. This system is designed to serve various types of users, including **Investors, Brokers, Analysts, Foreign Institutional Investors (FII), Domestic Institutional Investors (DII), and Unregistered Users**, by providing efficient stock market data management and analysis. InvestHUB offers key functionalities such as **stock tracking, trade execution, portfolio management, market analysis, and management of institutional investments**. It enables users to monitor stock performance, execute trades, analyze market trends, and manage investment portfolios. The specific use cases for each user type are detailed in the next section.

DATABASE OBJECTIVES

1. Investor-Specific Objectives

- **Trade Stocks:** Buy and sell stocks.
- **Portfolio Management:** Track and optimize your investments.
- **Custom Watchlists:** Create, edit, and organize watchlists for better market tracking.
- **Profit & Loss Reports:** Generate detailed reports to access financial performance.
- **Fund Management:** Add, withdraw, and allocate funds with ease.
- **IPO Insights:** Get the latest updates on upcoming and ongoing IPOs.
- **Dividend Tracker:** Access detailed dividend information for your investments.
- **Transaction Processing** – transactions status tracking.
- **Stock Performance Insights** – Store and analyze P/E ratio, EPS, and dividend yield for fundamental analysis.
- **Mutual Fund Investment Management** - Allow users to buy and sell mutual fund units seamlessly.
- **Mutual Fund Holdings & Stock Allocation Analysis** - Show which stocks a mutual fund holds and their percentage allocation.
- **Fund Performance & Risk Analysis** - Compare mutual funds based on returns, risk level, and expense ratio.

DATABASE OBJECTIVES

2. Investor-Specific Objectives

- **Transaction Analysis:** Monitor and manage user transactions across the platform.
- **Institutional Trade Tracking:** Track FII and DII trades, information for analysis.
- **IPO Insights:** Provide admins with the latest IPO information.
- **Block Deals Information:** Administer and store information on large institutional trades to ensure market transparency and compliance.

3. Analyst-Specific Objectives

- **Market Analysis Tools:** Provide prediction tools for market trends (help analysts keep up with stock performance, volatility, etc.).
- **Stock Performance Insights:** Perform in-depth analysis of stock performance, focusing on key metrics like P/E ratio, EPS, and dividend yield.
- **Institutional Trade Tracking:** Analyze institutional trade activities (FII/DII) to understand market movements.

DATABASE OBJECTIVES

- **Mutual Fund Holdings & Stock Allocation Analysis:** Analyze mutual fund portfolios, the stocks they hold, and their respective allocations to assess trends.
- **Fund Performance & Risk Analysis:** Assess the performance and risk factors of mutual funds in comparison to others, and provide recommendations to investors.

4. Foreign Institutional Investor (FII) and Domestic Institutional Investors (DII)

- **Institutional Trade Tracking** – Record and analyze FII and DII trading activities to assess market sentiment.
- **Block Deals Information** – Maintain records of large-volume trades between institutional investors for better market transparency.
- **Stock Performance Insights** – Store and analyze P/E ratio, EPS, and dividend yield for fundamental analysis.

TENTATIVE QUERIES

- Search the stock based on its **stock_id** or **stock_symbol**.
- Get available funds from a given user.
- Display today's **market trend** of a given stock.
- List the **upcoming IPO** details.
- Give **Mutual Funds' minimum investment** and Risk level.
- Give the **return of the specified mutual fund**.
- Get stock's **P/E ratio, EPS, Volume, 52-week Low, and High** based on its stock_id.
- List stocks which have **Market Capitalization > 20,000** and **P/E ratio < 50**.
- Select a stock and **buy a specified quantity** of it.
- Give **Total profit/loss** of the day.
- Stock **returns in the last year**.
- List **top 3 performing stocks** of a given **watchlist**.
- Give **which sector is best performing** in terms of **profit/loss** on a given date.
- List **stocks added to the portfolio from the watchlist**.
- Give **dividend value** for a given **sector, company-wise**.
- Transactions done by **FII's and DII's** on a given date.
- Which **block deals** happened during a given time period (8:45 to 9:00 or 2:05 to 2:20).
- List the **Total Dividends Paid** by a given company in the **Last Year**.

TENTATIVE QUERIES

- List the **Top 3 Performing Stocks** by Market Capitalization in the Last Year.
- Identify the **Most Active Users in Buying Stocks** (based on the number of buys).
- Get the **Sector-Wise Total Investment** from **Foreign Institutional Investors (FII)**.
- Find the **Total Investment in Mutual Funds** by each user.
- Get the **Top 5 Stocks Based on Dividend Yield** in the **Last Year**.
- List users who have a **portfolio with stocks from more than 2 different sectors**.
- Find the **Total Number of Block Deals** for each stock in the **Last Month**.
- Get the **Stocks That Were Added to the Watchlist** by more than 5 users.
- Identify the **Most Profitable Sector** for Investors in Terms of **Profit/Loss** in the **Last Year**.
- Get the **Total Block Deal Value** (by FII or DII) in a **Specific Date Range** for a specific stock.
- List **Mutual Funds with the Highest AUM (Asset Under Management)** in each **Risk Level Category**.
- Calculate the **Total Transaction Fee Collected by Brokers** in the **Last Month**.

TENTATIVE QUERIES

- **Set Stop-Loss** for a Given Stock at **5%**. That is, sell the stock if its value drops more than 5% below its average buy price.
- **Set a Target Price of ₹500** for a given stock and **purchase 200 quantities** of that stock automatically.
- **Total Investment by Sector** for Each Investor (grouped by sector).